

1 Part 4: Constructing Estimators

2 Summary

3 Recall that typically a probability model is **incompletely specified**. I.e.,
4 a particular model is assumed, but there are still unknown components,
5 often real-valued **parameters**. We use available data (a **sample**) to estimate
6 these parameters.

7 In this Part we describe methods for constructing estimators for unknown
8 parameters.

1 Terminology Review

2 The **parameters** are the constants that distinguish probability models in the
3 same “family.” Examples of “families” include “normal,” “exponential,”
4 and so forth.

5 A **statistic** is a function of the RVs being used to model the observable data.
6 Examples include the sample mean, the sample variance, etc. A statistic is
7 a **random variable**.

8 An **estimator** is a statistic which is used to estimate an unknown parame-
9 ter. It is also a **random variable**.

10 The “hat notation” is standard for denoting an estimator: $\hat{\beta}$ is an estimator
11 for β , for example. θ is often used to denote a generic parameter, $\hat{\theta}$ is its
12 estimator.

- 1 An estimator $\hat{\theta}$ is said to be **an unbiased estimator for θ** if $E(\hat{\theta}) = \theta$.
- 2 The variability in an estimator (its precision) is naturally quantified by its
- 3 variance, $V(\hat{\theta})$.
- 4 The **standard error (SE) of $\hat{\theta}$** is the square root of its variance. The SE is of-
- 5 ten reported along with an estimate in order to give a sense of the amount
- 6 of error one should anticipate in that estimate.

1 The Method of Moments

2 Simple cases such as estimating the population mean and estimating the
3 population proportion can be misleading: In those cases it is “obvious”
4 what the natural estimator is, since one can use the sample mean or sample
5 proportion as the estimator, respectively.

6 In other cases, the choice of estimator is not so clear.

7 We first consider a simple approach to constructing estimators, the **method**
8 **of moments**.

1 First, recall that for any positive integer k , the quantity $E(X^k)$ is called the
2 k^{th} moment for the distribution of X . We might call this the “ k^{th} population
3 moment” to be more clear.

4 Imagine that I have random variables $X_1, X_2, \dots, X_n$ that model the ob-
5 servable sample. Then the k^{th} sample moment is

6
$$M_k = \sum_{i=1}^n X_i^k / n$$

7 Note that M_k is our best estimator for $E(X^k)$, for instance $E(M_k) = E(X^k)$

1 Now, suppose that $X_1, X_2, \dots, X_n$ are i.i.d. each with density $f_X(x; \boldsymbol{\theta})$.

2 Here, $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_p)$ is a vector of parameters.

3 Set up a system of p equations, setting the first p sample moments equal to
4 the first p population moments:

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$$M_1 = E(X)$$

6
$$M_2 = E(X^2)$$

7
$$\vdots \quad \vdots$$

8
$$M_p = E(X^p)$$

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10 We note that the sample moments are all quantities we can determine from
11 the available sample. We also note that the population moments are func-
12 tions of the unknown parameters.

1 We can now solve this system of equations for $\theta_1, \theta_2, \dots, \theta_p$:

2
$$\theta_1 = g_1(X_1, X_2, \dots, X_n)$$

3
$$\theta_2 = g_2(X_1, X_2, \dots, X_n)$$

4
$$\vdots \quad \quad \quad \vdots$$

5
$$\theta_p = g_p(X_1, X_2, \dots, X_n)$$

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7 These equations are built on the (incorrect) belief that $M_i = E(X^i)$.

8 In fact, $M_i \approx E(X^i)$, so

9
$$\theta_i \approx g_i(X_1, X_2, \dots, X_n)$$

10 and we construct our estimator as

11
$$\hat{\theta}_{i,\text{MOM}} = g_i(X_1, X_2, \dots, X_n)$$

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. from the Exponential(λ)
2 distribution. (Recall that this means that $E(X_i) = 1/\lambda$.) What is the method
3 of moments estimator for λ ?

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- 1 **Exercise:** What is the result if, instead of the first moment, the second mo-
2 ment is used to construct the estimator for λ in the previous example?

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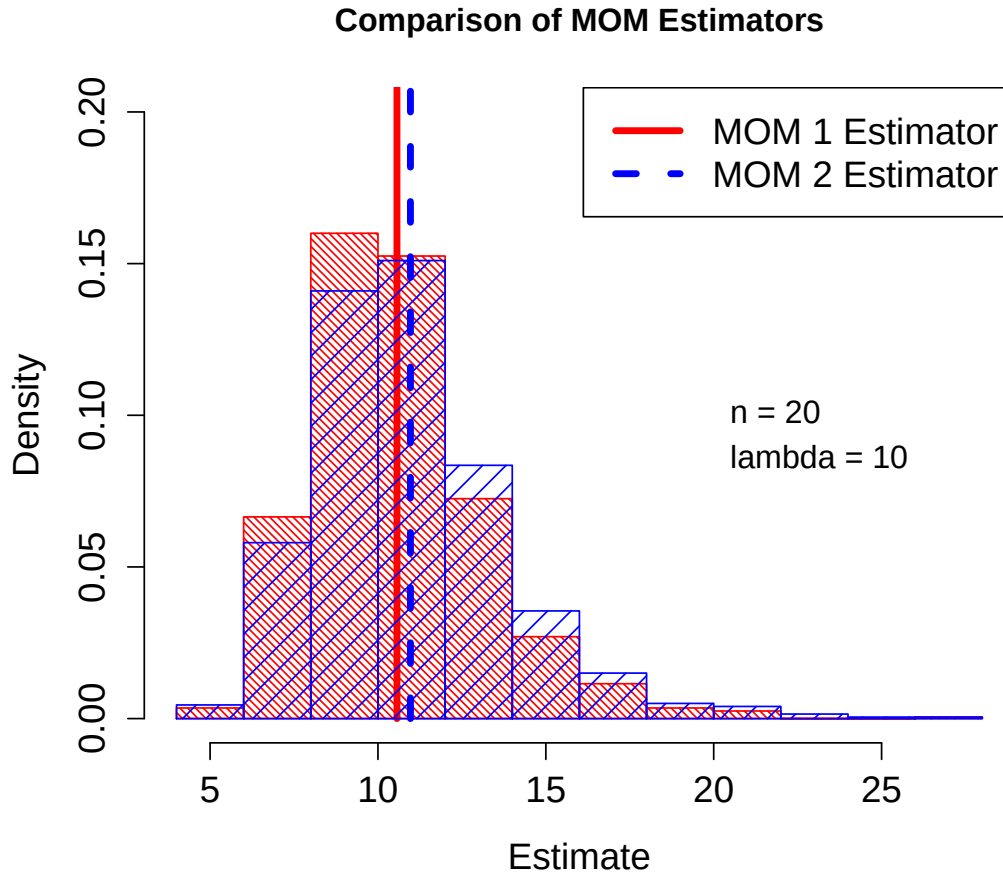
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1 **Exercise:** Is the method of moments estimator found above (using the first
2 moment) an unbiased estimator for λ ? Can you adjust it to make it an
3 unbiased estimator?

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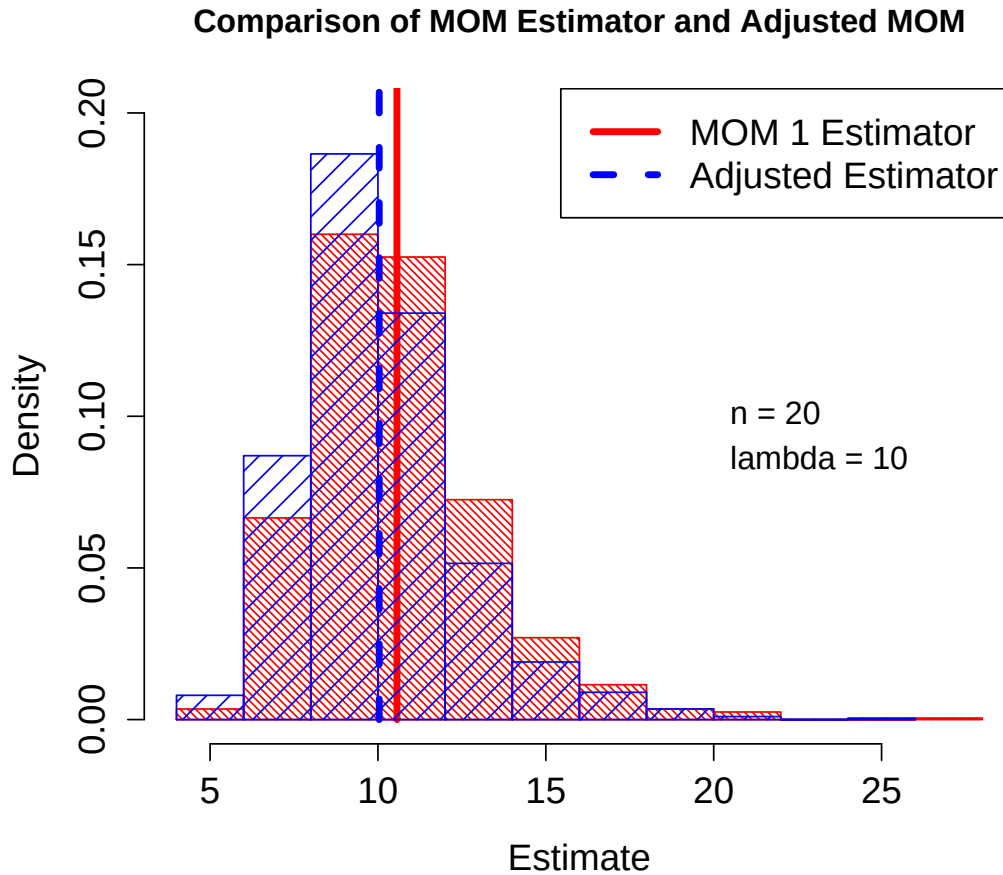
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- 1 The comparisons made above between the three estimators are based **en-**
2 **tirely on simulations.** There is **no** observed data involved at all. This is
3 the **frequentist approach** to assessing estimators: **how does the estimator**
4 **perform in repeated use?**
- 5 As seen above, simulations are a useful tool for cases where the mean and
6 variance derivations are not available.

- 1 **Exercise:** Interpret the output below. What is the approximate MSE of each
2 of the three estimators?

```
> print(c(mean(mom1), mean(mom2), mean(adju)))  
[1] 10.56553 10.96759 10.03725
```

```
> print(c(var(mom1), var(mom2), var(adju)))  
[1] 6.574420 8.224773 5.933414
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1 Maximum Likelihood Estimation

2 **Maximum likelihood estimation (MLE)** is a powerful, general approach to
3 constructing estimators.

4 We will begin the explanations with simple, one-dimensional examples,
5 but the concepts extend to complex, high-dimensional models in a natural
6 manner.

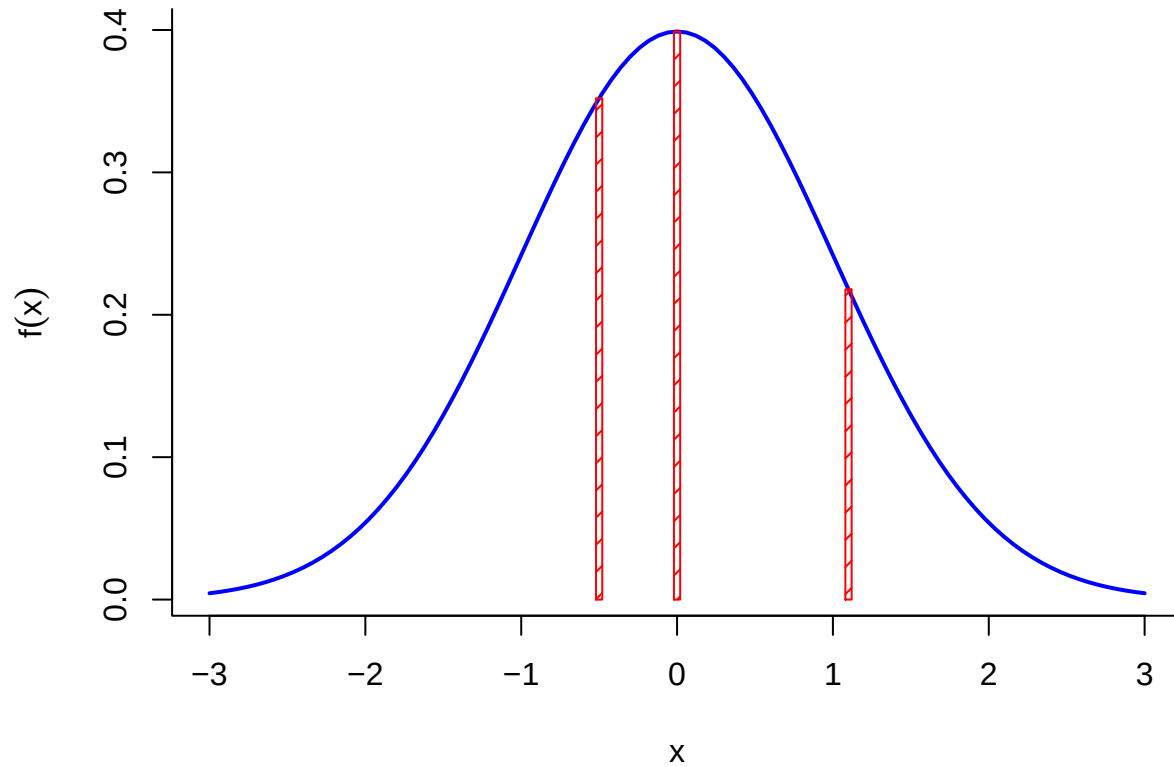
7 Importantly, there is also a general way of calculating standard errors of
8 MLEs, and, under a wide range of conditions, the MLE is approximately
9 normally distributed. This makes inference with MLEs straightforward.

1 Basic Setup

2 Recall that if X is a continuous random variable with density f_X , that does
3 **not** mean that $P(X = a) = f_X(a)$, i.e., the density does **not** return proba-
4 bilities. Indeed, $P(X = a) = 0$ for all a .

5 **It is true** that the density is higher in regions where there is the most
6 probability: For fixed, small $\epsilon > 0$, if we want to find a to maximize
7 $P(a - \epsilon \leq X \leq a + \epsilon)$, then we would look for the a that maximizes $f_X(a)$.

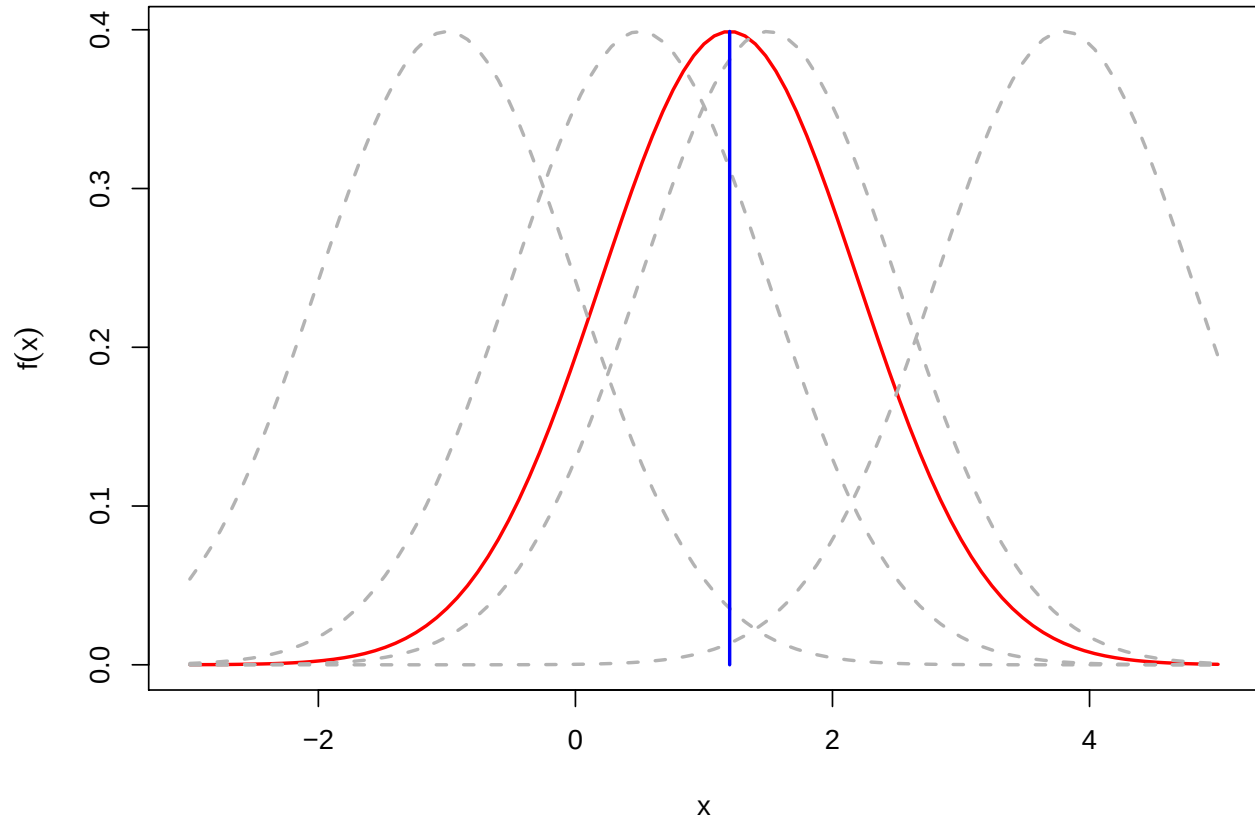
8 See the figure on the following slide. The standard normal density is de-
9 picted. The most likely values are close to zero, where the density is maxi-
10 mized.



As previously stated, we use probability distributions such as the normal to “model” the real processes that generate data. So, if we assume that our observable data is drawn from the standard normal distribution, the density above specifies where we are more or less likely to see our data value.

We can turn this idea around: Suppose that we observe data $x = 1.2$, and we are willing to assume that it came from a normal distribution with mean μ and variance one. What value of μ makes the observation $x = 1.2$ the most likely?

See the figure on the following slide. Different choices of μ are displayed. It's not difficult to conclude that when $\mu = 1.2$, the **likelihood** is maximized.



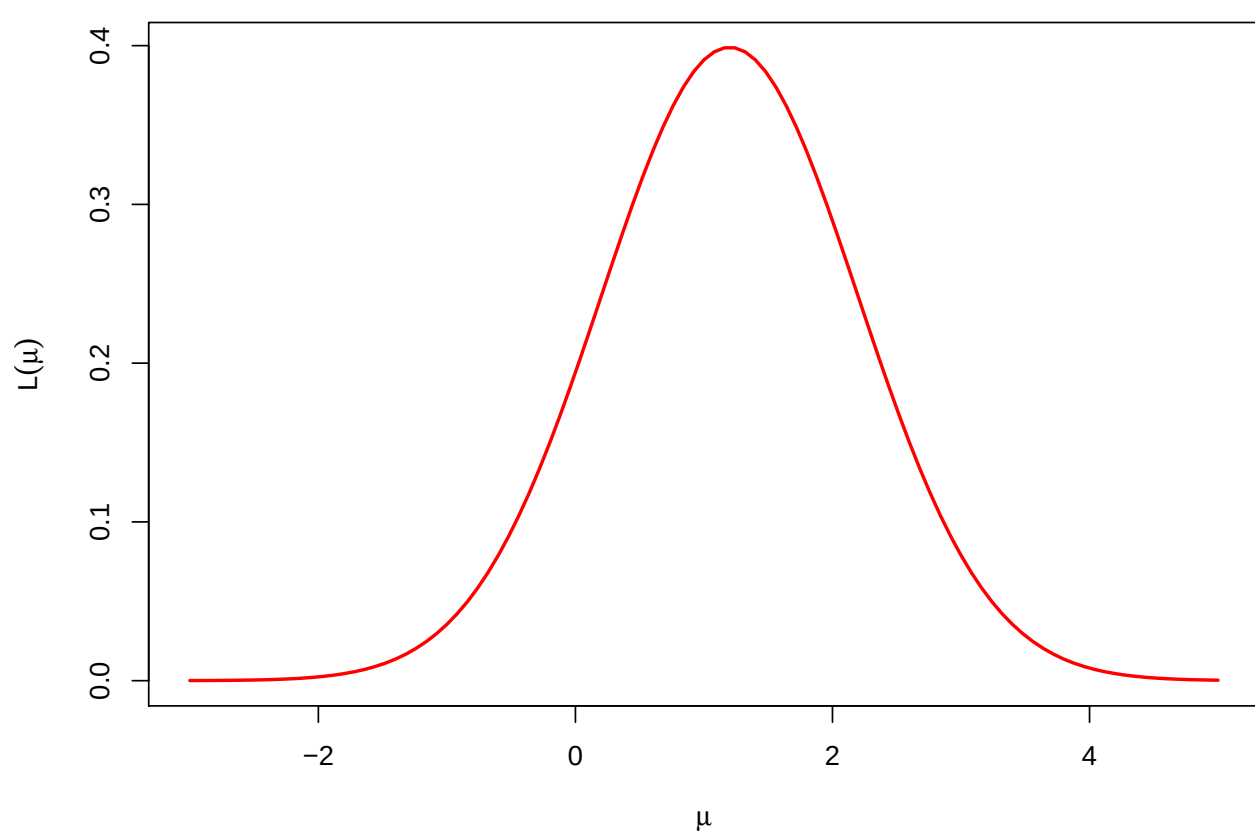
1 The **likelihood function** is constructed by calculating the distribution for
2 the data, and then **holding the data constant** and allowing the parame-
3 ter(s) to vary.

4 In the simple example above, we assume that we observe a single data
5 point x that was generated from a normal distribution with mean μ and
6 variance one. Hence, the distribution is

7
$$f(x; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2}\right)$$

8 If we fix the data $x = 1.2$, and think of this as a function of μ , then we arrive
9 at our likelihood function:

10
$$L(\mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(1.2 - \mu)^2}{2}\right)$$



- 1 The **maximum likelihood estimator** for the parameter is the choice of the
2 parameter which maximizes the likelihood function.
- 3 Heuristically, we ask: Among all possible choices for the parameter, which
4 one makes our observed data the most likely?
- 5 It is worth stressing that the likelihood function is **not** itself a probability
6 distribution. It is **not** specifying the relative “probabilities” for different
7 values of the parameter.
- 8 Remember that in the frequentist approach, the unknown parameter is a
9 constant, and not random.

Exercise: An urn contains black and white balls. All that we know is that the balls have a color ratio of 1 to 3. (We do not know if there are more white or more black balls.) We draw three balls **with replacement** from the urn. Let X equal the number of black draws. Derive the likelihood function and the MLE as a function of X .

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Exercise: Now suppose that θ equals the proportion of balls in the urn which are black. We know that $0 < \theta < 1$. We draw ten times with replacement, and count seven black balls and three white balls. Derive the MLE for θ .

1 The preceding examples involved a single, observable random variable. In
2 most cases, we have a sequence of random variables in the model.

3 A common case is where $X_1, X_2, \dots, X_n$ are i.i.d. with density $f_X(x; \theta)$,
4 where θ is the parameter to be estimated. In this case, the joint distribution
5 of the random variables is

6
$$\prod_{i=1}^n f_{X_i}(x_i; \theta)$$

7 and the likelihood function is

8
$$L(\theta) = \prod_{i=1}^n f_{X_i}(x_i; \theta)$$

1 Finding the MLE Analytically

2 In some situations the MLE can be derived analytically, usually via calcu-
3 lus, but sometimes via direct inspection of the likelihood.

4 One “trick” that is often very helpful: The likelihood function is positive
5 at its maximum. Hence, the likelihood is maximized at the same θ as does
6 the logarithm of the likelihood.

7 Hence, we often work with the **log likelihood** instead of the likelihood.
8 This is often denoted $\ell(\theta) = \log L(\theta)$.

9 In the case where $X_1, X_2, \dots, X_n$ are i.i.d. $f_X(x; \theta)$, then

10
$$\ell(\theta) = \log \prod_{i=1}^n f_{X_i}(x_i; \theta) = \sum_{i=1}^n \log f_{X_i}(x_i; \theta)$$

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. with the Exponential(λ)
2 distribution. Derive the MLE for λ .

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- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. with the $\text{Beta}(\alpha, 1)$ distri-
2 bution. Derive the MLE for α . Compare this with the MOM estimator.

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1 **Exercise:** Assume that $X_1, X_2, \dots, X_n$ are i.i.d. with the Normal(μ, σ^2) dis-
2 tribution. Derive the MLE for μ and σ^2 .

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1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Uniform}(0, \theta)$. Derive the
2 MLE for θ . What is the MOM in this case?

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